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Times Square Sampling: An Adaptive Algorithm for Free Energy Estimation

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ABSTRACT

Estimating free energy differences, an important problem in computational drug discovery and in a wide range of other application areas, commonly involves a computationally intensive process of sampling a family of high-dimensional probability distributions and a procedure for computing estimates based on those samples. The variance of the free energy estimate of interest typically depends strongly on how the total computational resources available for sampling are divided among the distributions, but determining an efficient allocation is difficult without sampling the distributions. Here we introduce the *Times Square sampling* algorithm, a novel on-the-fly estimation method that dynamically allocates resources in such a way as to significantly accelerate the estimation of free energies and other observables, while providing rigorous convergence guarantees for the estimators. We also show that it is possible, surprisingly, for on-the-fly free energy estimation to achieve lower asymptotic variance than the maximum-likelihood estimator MBAR, raising the prospect that on-the-fly estimation could reduce variance in a variety of other statistical applications. Supplementary materials for this article are available online.

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1. Introduction

An outstanding problem in computational statistics is how to efficiently sample multi-modal, high-dimensional probability distributions in order to estimate ratios of their normalizing constants. One important instance of this problem is the calculation of free energy differences, which has applications in various fields, including computational drug discovery. The probability distributions to be sampled in order to estimate free energy differences are Gibbs distributions, which give the probability that a physical system adopts a certain state; the free energy difference between a pair of Gibbs distributions is proportional to the logarithm of the ratio of their normalizing constants. Sampling Gibbs distributions is typically computationally intensive, and often intractable with existing methods.

Sampling algorithms such as Markov chain Monte Carlo (MCMC) and molecular dynamics (MD) explore the state space of a simulated system through local moves and can get trapped in regions of high probability density (which are also called “modes”). To facilitate sampling of the many modes, techniques such as replica exchange (Swendsen and Wang 1986; Geyer 1991; Sugita and Okamoto 1999) and simulated tempering (Marinari and Parisi 1992; Geyer 1994) are commonly used. These methods “enhance” the sampling by introducing a parameterized family of overlapping distributions, and the problem becomes one of sampling each distribution within the family, which is typically still a formidable computational task. If enough uncorrelated samples can be collected, free energy differences within the family can be estimated using a technique such as the weighted histogram analysis method (WHAM) (Bouzida, Kumar, and Swendsen 1992), or its “binless” version,

the multistate Bennett acceptance ratio (MBAR) (Geyer 1994; Meng and Wong 1996; Kong et al. 2003; Shirts and Chodera 2008).

A major drawback of enhanced sampling methods such as replica exchange and simulated tempering is that their sampling efficiency is strongly dependent on the underlying free energy landscape, which is, by the nature of the estimation problem, typically unknown a priori. Each distribution to be sampled does not contribute equally to the variance (i.e., the mean-squared error) of the free energy difference estimate of interest, and appropriately allocating limited sampling resources among these distributions requires that their relative contributions be estimated. These estimates are, however, themselves determined by underlying samples, creating a cycle that makes it difficult to improve the efficiency of the free energy estimation procedure.

To improve the efficiency of free energy estimation, methods that blend sampling and estimation have been developed in recent years, in which on-the-fly estimates guide sampling. Using the method of stochastic approximation, Z. Tan has shown that one can use simulated tempering to estimate free energy differences on the fly, and moreover do so with the smallest possible asymptotic variance (i.e., in the limit of infinitely many samples) when the allocation of resources is fixed (Tan 2017). A related algorithm, accelerated weighted histogram (AWH), estimates on the fly not only free energy differences, but also ensemble averages for other functions defined on the state space (i.e., observables) (Lindahl, Lidmar, and Hess 2014; Lidmar 2012). This allows the sampler to estimate an asymptotic allocation of resources that lowers the variance of the desired free energy difference estimate (Lindahl, Lidmar, and Hess 2014).

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These advances underline the value of incorporating estimate information into the sampling procedure in order to reduce the computational effort spent sampling less relevant parts of the free energy landscape (Andrieu and Moulines 2006; Atchadé and Rosenthal 2005; Atchadé et al. 2011; and Haario, Saksman, and Tamminen 2001). Without establishing explicit conditions that guarantee convergence of free energy estimators, however, it is difficult to develop resource allocation schemes that can further reduce variance or accelerate convergence.

Here we present Times Square sampling (TSS), an algorithm that provides a straightforward recipe to estimate ensemble averages on the fly for any function defined on the state space, and enables the use of this information to allocate computational resources to an intermediate distribution within the parameterized family where more sampling is needed. TSS gives simple conditions that establish how resources can be allocated based on the state of the system, the parameter of the intermediate distribution being sampled, and the current free energy estimates, while guaranteeing convergence. This enabled us to introduce, for example, a *visit-control mechanism*, which modulates the likelihood of visiting upcoming distributions based on whether they have been under- or over-visited, that in our experiments greatly sped up convergence of the free energy estimates.

Importantly, we also prove that an additional property of TSS that we observed, “self-adjustment” (a term coined by Z. Tan for a property by which the likelihood of sampling upcoming distributions is modulated based on the current free energy estimates (Tan 2017)), makes it possible for on-the-fly estimation algorithms to achieve lower asymptotic variance than the maximum-likelihood estimator MBAR (Vardi 1985; Kong et al. 2003). Self-adjustment enables this improvement—a surprising one, given that maximum-likelihood estimators have the lowest variance when independent samples are available (Ibragimov and Has’minskii 1981; Gill, Vardi, and Wellner 1988; Tan 2004; Rao 1945)—by taking advantage of on-the-fly information, which is unavailable to MBAR. This raises the intriguing prospect that it may be possible to develop on-the-fly estimation procedures that improve over maximum-likelihood estimation in a variety of other statistical applications.

We also describe features of TSS relevant to its practical implementation. Performing energy evaluations for all intermediate distributions can be prohibitively expensive. We thus include a windowing system, which limits the number of energy evaluations required to update the estimates. Another issue is that convergence can be slow if the initial free energy estimates are sufficiently inaccurate. We thus introduce a history-forgetting mechanism to TSS, which in many cases significantly reduces the number of steps it takes before the statistical variance becomes the dominant contributor to the error. We summarize our implementation of TSS in Section 3 (and provide a full implementation in Part 2 of the supplementary materials), and in Section 4 we study its behavior using a numerical example.

2. Mathematical Background and Theoretical Derivation of TSS

Our starting point is a compact state space $\mathcal{S} \subset \mathbb{R}^N$, elements of which are (for instance) positions of atoms; a compact param-

eter space $\Lambda \subset \mathbb{R}^d$, which represents a quantity to be varied (such as temperature); and a family of probability distributions $\{\rho_\lambda(x) dx, \lambda \in \Lambda\}$ defined on $\mathcal{S}$ and parameterized by $\lambda \in \Lambda \subset \mathbb{R}^d$. The distributions are defined in terms of dimensionless potential functions $H_\lambda : \mathbb{R}^N \rightarrow \mathbb{R}$ (which we will call *Hamiltonians*). These functions are assumed to be continuously differentiable in (x, λ) , and they specify the density $\rho_\lambda(x)$ with respect to the Lebesgue measure dx by

$$\rho_\lambda(x) = e^{-H_\lambda(x)} / Z_\lambda^*, \quad x \in \mathcal{S}, \quad \lambda \in \Lambda, \quad (1)$$

where $Z_\lambda^* = \int_{\mathcal{S}} e^{-H_\lambda(x)} dx$ is a normalizing constant called the partition function. It is desired to estimate ratios of partition functions $Z_{\lambda'}^* / Z_\lambda^*$ or, equivalently, free energy differences $F_{\lambda'}^* - F_\lambda^*$, with $F_\lambda^* = -\log Z_\lambda^*$ defining the (dimensionless) free energy. In practice, we discretize $\Lambda \subset \mathbb{R}^d$ into points $\{\lambda_1, \dots, \lambda_K\} \subset \Lambda$, and going forward we will write H_k for H_{λ_k} , F_k^* for $F_{\lambda_k}^*$, etc.

In simulated tempering, the method on which TSS is built, one specifies a rung density $\gamma = (\gamma_1, \dots, \gamma_K)$ associated with the discretization $\{\lambda_1, \dots, \lambda_K\}$ and then samples the joint distribution $\gamma_k \rho_k(x) dx \mathcal{C}(dk)$, where $\mathcal{C}(dk)$ is the counting measure on the set $[K] = \{1, \dots, K\}$ (i.e., the measure that assigns a unit mass to every singleton set $\{k\} \subset [K]$). To avoid degeneracies, we fix some $0 < \varepsilon < 1$ and define $\mathcal{P}^\varepsilon(K)$ to be the set of probability densities ϕ on $[K]$ satisfying $\phi_k \geq \varepsilon$ for each $k \in [K]$, and we only consider rung densities $\gamma \in \mathcal{P}^\varepsilon(K)$.

Throughout, we assume that the space $\Lambda \subset \mathbb{R}^d$ is low-dimensional (e.g., $d \leq 5$), in contrast to the much larger dimensionality of the state space $\mathcal{S}$, which contains 10^4 – 10^7 degrees of freedom in typical chemical physics applications. Consequently, conditional Monte Carlo sampling of the Λ space (selecting λ_k given x) can be done more efficiently than sampling of the state space (selecting x given λ_k), a limiting situation we refer to as a *two-timescale sampling* scenario. Whereas $\{\rho_k, k \in [K]\}$ is assumed given, we note that γ is a free parameter that can be adjusted to improve the efficiency of the algorithm, since it prescribes the frequency with which computational resources are allocated to the sampling of any of the K distributions $\rho_k(x) dx$.

2.1. The Target Invariant Distribution and Stochastic Approximation

For any fixed $(\gamma, F) \in \mathcal{P}^\varepsilon(K) \times \mathbb{R}^K$, we seek to sample the target joint invariant density

$$p_{\gamma, F}(x, k) = \frac{\gamma_k e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}} \rho_k(x). \quad (2)$$

An expectation with respect to $p_{\gamma, F}(x, k) dx \mathcal{C}(dk)$ will be denoted by $\mathbb{E}_{\gamma, F}$, where we recall that $\mathcal{C}$ is the counting measure on $[K]$. The marginal densities of $p_{\gamma, F}(x, k)$ are

$$\begin{aligned} p_{\gamma, F}^{[K]}(k) &= \frac{\gamma_k e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}}, \\ p_{\gamma, F}^{\mathcal{S}}(x) &= \sum_{k \in [K]} \frac{\gamma_k e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}} \rho_k(x). \end{aligned} \quad (3)$$

We assume that for each $k \in [K]$ there is a transition kernel $\mathcal{T}_k(x'|x)$ having $\rho_k(x) dx$ as its unique invariant distribution, and we write $\mathcal{T}_k^{(n)}(x'|x)$ for its n -step iteration. The simulated tempering algorithm samples (2) for any fixed $(\gamma, F) \in \mathcal{P}^e(K) \times \mathbb{R}^K$ by composing the pair of Markov transition kernels with densities

$$p_{\gamma, F}^k(x', k|x, k) = \delta(x' - x) \frac{\gamma_k e^{F_k - H_k(x)}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - H_\ell(x)}}, \quad (4)$$

$$p_{\gamma, F}^x(x'', k'|x', k') = \delta(k'' - k') \mathcal{T}_{k'}^{(n)}(x''|x'). \quad (5)$$

The above densities are written with respect to the product $dx \mathcal{C}(dk)$ on $\mathcal{S} \times [K]$; δ denotes the Dirac density (the Kronecker symbol for a discrete distribution). We let $P_{\gamma, F}$ denote the composition $P_{\gamma, F}^x \circ P_{\gamma, F}^k$, and we assume that $\mathcal{T}_k(x'|x)$ is sufficiently regular that $P_{\gamma, F}$ is uniformly ergodic with stationary distribution $p_{\gamma, F}(x, k) dx \mathcal{C}(dk)$, an assumption that holds in practical examples such as molecular dynamics simulations.

The main difficulty in sampling the density (2) is that it requires a reasonably accurate guess of the free energies for good sampling. To remedy this, one can try to use stochastic approximation (see, e.g., Benveniste, Métivier, and Priouret 1990; Kushner and Yin 1997; Robbins and Monro 1951; Kushner and Yin 2003; Liang 2010; Liang, Liu, and Carroll 2007; Liang, Wu, and Song 2014) to estimate the free energy differences on the fly. Briefly, stochastic approximation can be used to estimate the root of an equation $g(\theta) \equiv \int G(y, \theta) p_\theta(y) dy = 0$ using a general recursion of the form

$$\theta^{t+1} = \theta^t + \frac{1}{t+1} \mathbf{\Gamma} G(\mathcal{Y}^{t+1}; \theta^t), \quad (6)$$

where $\{\mathcal{Y}^t, t \geq 0\}$ is a stochastic process generated by drawing $\mathcal{Y}^{t+1}$ from $P_{\theta^t}(\mathcal{Y}^t, \cdot)$, $P_\theta(y, \cdot)$ is an ergodic transition kernel with stationary density $p_\theta(y)$, and $\mathbf{\Gamma}$ is a matrix called the “gain matrix.” The function $G(y; \theta)$ is called the “observable,” and $g(\theta)$ is called the “mean field.”

With $p_{\gamma, F}(k|x)$ denoting the conditional density $\frac{p_{\gamma, F}(x, k)}{p_{\gamma, F}(x)}$, one possible recursion for the free energy estimates is

$$\begin{aligned} F_k^{t+1} &= F_k^t + \frac{1}{t+1} G_k^F(\mathcal{X}^{t+1}; F^t) \\ &= F_k^t + \frac{1}{t+1} \left(1 - \frac{1}{\gamma_k} p_{\gamma, F^t}(k|\mathcal{X}^{t+1}) \right) \\ &= F_k^t + \frac{1}{t+1} \left(1 - \frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} \right), \end{aligned} \quad (7)$$

which was introduced in Self-Adjusted Mixture Sampling (SAMS) (Tan 2017) and AWH (Lindahl, Lidmar, and Hess 2014). This stochastic approximation recursion has the mean field

$$g_k^F(F) = \mathbb{E}_{\gamma, F}[G_k^F(\mathcal{X}; F)] = 1 - \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}}, \quad \forall k \in [K], \quad (8)$$

and the root of the above equation is $F = F^*$. Notably, Tan has shown that (7) is presented with its optimal gain matrix $\mathbf{\Gamma} = I_{K \times K}$; that is, that the negative inverse of the Jacobian of $g_k^F(F)$ evaluated at $F = F^*$ is equal to the identity matrix.

2.2. The Times Square Sampling Algorithm

Although the stochastic recursion (7) in theory allows one to estimate free energy differences on the fly, by itself it is of limited practical value for chemical physics applications. There are two main sources of difficulty encountered in such applications: (i) the emergence of a critical slowdown in convergence whenever the parameter space $\mathbf{\Lambda}$ is large enough that the variability in the free energies across $\mathbf{\Lambda}$ make an accurate initial guess infeasible, which we illustrate by means of a Gaussian model in Section 4, and (ii) the vastness of the space $\mathcal{S}$ in typical chemical physics applications, which makes it strongly desirable to apply computational resources as effectively as possible to sampling the joint density $p_{\gamma, F}(x, \lambda)$. In particular, it is important to keep the estimation of free energies efficient by judiciously allocating resources among the rungs. Furthermore, the estimation of free energies and sampling are generally done on different machines, leading to communication latency that, for adaptive methods, creates a synchronization expense that should ideally be mitigated and/or hidden.

In this section we partially address the critical slowdown problem by introducing *visit control*, which is shown in Section 4 to improve the convergence (7) on the Gaussian model. In Section 3, we introduce a *windowing system*, which prevents the critical slowdown by decomposing the single recursion (7) into a collection of weakly coupled recursions restricted to overlapping subdomains of $\mathbf{\Lambda}$. We emphasize that removing either of these two aspects of the TSS algorithm may result in a failure to converge in large-scale simulations (as seen in the numerical example, Section 4). Additionally, in order to determine a judicious allocation of computational resources among the rungs, in Section 2.2.2 we provide a way of estimating averages of general observables, and in Section 2.2.3 we describe a property of the TSS algorithm called *self-adjustment* that allows one to partially overcome the synchronization expense.

2.2.1. Visit Control

The main reason for the slow convergence of the recursion (7) is that the observable G_k^F is bounded above by 1. If at some time τ we have $F_k^\tau \ll F_k^*$, then the inequality $F_k^\tau - F_k^* \leq \sum_{s=\tau+1}^t s^{-1} = O(\log t)$ and the slow divergence of the harmonic series implies that F_k^τ requires exponentially many steps to increase to within a small neighborhood of F_k^* . It has been noted that a way of handling this difficulty is by increasing the gain factor from t^{-1} , typically to $t^{-\beta}$ for some $\beta \in (0.5, 1]$ (Tan 2017). With this approach, not only is the rate of decay of the root-mean-square error slower than the celebrated $t^{-1/2}$, but the approach is unattractive for simulations in which the samples in $\mathcal{S}$ have long correlation times. Some methods propose adapting the gain—either deterministically, as in Tan (2017) (use $\beta < 1$ for $t < t_0$ and $\beta = 1$ for $t \geq t_0$), or stochastically, as in Kesten’s algorithm (Kesten 1958; Delyon and Juditsky 1993) (decreasing the step size only if the estimates seem to fluctuate near the equilibrium). These modifications are powerful enough for many applications, but are limited by the assumption that the gain is nonincreasing. In chemical physics applications, it is often the case that the sampler must jump between multiple modes of the distribution with long residence times between jumps, and thus the estimates

must quickly respond to changes in x even long after F^t appears to have converged.

Rather than modifying the gain factor t^{-1} , TSS uses its visit control mechanism, which has the effect of modifying the magnitude of the mean field when F is far away from its exact value F^* . To proceed, let ϵ_π and ϵ_γ be defined such that $0 < \epsilon_\pi \ll \epsilon_\gamma < 1$, and let $\gamma \in \mathcal{P}^{\epsilon_\gamma/K}(K)$. In brief, the role of ϵ_γ is to guarantee that each state $k \in [K]$ will be appropriately visited asymptotically (which will be used in 1.2.2), whereas the role of ϵ_π is to transiently modify the mean field g_k^F (8) to allow for a sufficiently large gain, up to $\mathcal{O}(\epsilon_\pi^{-1})$, to compensate for the small $\mathcal{O}(t^{-1})$ learning rate without ruining the asymptotic $t^{-1/2}$ behavior of the errors.

Visit control is motivated by the observation that when F^t is far away from F^* and t is moderately large, the update (7) is small enough that F^t is essentially fixed. If we freeze F and sample from $p_{\gamma,F}$, the marginal rung occupancies

$$\frac{1}{t} \sum_{s=1}^t 1_{\{k\}}(K^s) \approx \frac{\gamma_k e^{F_k^* - F_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell^* - F_\ell}} \quad (9)$$

are exponentially tilted away from the desired reference γ . Formally, we could fix this problem by replacing γ with the rung distribution π° defined by

$$\pi_k^\circ(F, \gamma) = \frac{\gamma_k e^{F_k^* - F_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell^* - F_\ell}}. \quad (10)$$

Of course, π° is not available, since it assumes knowledge of the exact free energies F^* , but to avoid the exponential tilting away from γ , it is important to at least approximate π° . We do so by introducing the *tilts*

$$o_k^t = \frac{1}{t} \sum_{s=1}^t \frac{1_{\{k\}}(K^s)}{\gamma_k}, \quad (11)$$

and a continuously differentiable function $\pi(F, \gamma, o) : \mathbb{R}^K \times \mathcal{P}^\varepsilon(K) \times \mathbb{R}_{>0}^K \rightarrow \mathcal{P}^\varepsilon(K)$ for substitution in (4) and (7) in place of γ . If we assume that $o_k \propto e^{F_k - F_k^*}$ (as in (9)), an approximation for π_k° is

$$\pi_k(F, \gamma, o) \approx \frac{\frac{\gamma_k}{o_k}}{\sum_{\ell \in [K]} \frac{\gamma_\ell}{o_\ell}}. \quad (12)$$

More generally, TSS allows $\pi \in \mathcal{P}^\varepsilon(K)$ to be a function of any stochastic estimate (not necessarily o), all while preserving convergence guarantees on F^t (see Theorem 1 in the supplementary materials). This generality enables the user to use on-the-fly estimates to determine a rung distribution that is suited for a particular application. For the purpose of mitigating the slow convergence of the free energy estimates, we improve the mixing in $[K]$ by approximating π° using the tilts o^t , creating a temporary bias in the rung density that disappears asymptotically. There may be many (yet unknown) useful functions π ; the Self-Healing Umbrella Sampling algorithm (Fort et al. 2017), for instance, is reminiscent of using $\pi_k \propto \gamma_k / \theta_k$ for some estimate θ_k analogous to the tilts o_k .

For the remainder of this section, we focus on the use of o^t (as defined in (11)) to estimate π° . In particular, we cast this problem as finding the root of a steady-state equation, an

approach that will be important in Section 3 when we extend the definition of visit control to the windowing system. Equation (11) can be written as a stochastic approximation estimator in terms of the observables

$$G_k^o(j; F, o) = \frac{1}{\gamma_k} 1_{\{k\}}(j) - o_k, \quad (13)$$

which have the mean field

$$\begin{aligned} g_k^o(F, o) &= \mathbb{E}_{\pi, F} \left[\frac{1}{\gamma_k} 1_{\{k\}}(K) - o_k \right] = \frac{1}{\gamma_k} p_{\pi, F}^{[K]}(k) - o_k \\ &= \frac{\gamma_k^{-1} \pi_k(F, \gamma, o) e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell(F, \gamma, o) e^{F_\ell - F_\ell^*}} - o_k. \end{aligned} \quad (14)$$

Based on the two-timescale separation mentioned in Section 2, we assume $\dot{o}_k(t) = g_k^o(F, o(t))$ evolves at its steady state in o for any fixed F , and we view the relation $o_k \propto e^{F_k - F_k^*}$ as a consequence of the stationarity condition $g_k^o(F, o) = 0$ for $\pi_k = \gamma_k$. More generally, we assume π is chosen so that there is a unique $o^*(F)$ satisfying $g_k^o(F, o) = 0$. In Section 3, we will use the relation $g_k^o(F, o) = 0$ to define an auxiliary estimate $F^\circ(F, o)$ of the free energies F^* , and we will assume this estimate satisfies $F^\circ = F^*$ if and only if $o = o^*(F)$.

For any $\eta > 0$, we define visit control as the choice

$$\pi_k^{\text{TSS}}(F, \gamma, o) = \frac{\gamma_k / o_k^\eta}{\sum_{\ell \in [K]} \gamma_\ell / o_\ell^\eta}, \quad (15)$$

which can be expressed in terms of F° as

$$\pi_k^{\text{TSS}}(F, \gamma, o) = \frac{\gamma_k e^{(F_k^\circ - F_k)\eta/(\eta+1)}}{\sum_{\ell=1}^K \gamma_\ell e^{(F_\ell^\circ - F_\ell)\eta/(\eta+1)}}. \quad (16)$$

We call η the visit control parameter. The case $\eta = 0$ corresponds to no visit control, whereas the other extreme $\eta = \infty$ corresponds to “infinitely strong” visit control, for which π^{TSS} is formally equal to π° , but which is unattainable, as it would require $o_k = 1$ because of (15). In practice we use $\eta \in (0, \infty)$, with larger η creating a stronger bias to under-visited rungs. Additionally, we redefine $\pi_k^{\text{TSS}}(F, \gamma, o)$ through the ϵ_π -regularization

$$\pi_k^{\text{TSS}}(F, \gamma, o) := (1 - \epsilon_\pi) \pi_k^{\text{TSS}}(F, \gamma, o) + \epsilon_\pi \gamma_k, \quad (17)$$

where the symbol $\pi_k^{\text{TSS}}(F, \gamma, o)$ appearing in the right-hand side of (17) denotes the right-hand side of (16). With this choice, the ϵ_π -regularization is certain to leave γ_k unchanged if applied to it. We note that

$$\pi_k^{\text{TSS}}(F, \gamma, o) \geq \epsilon_\pi \gamma_k \geq \epsilon_\pi \epsilon_\gamma K^{-1}, \quad (18)$$

so $\pi^{\text{TSS}}(F, \gamma, o) \in \mathcal{P}^\varepsilon(K)$ for all $(F, o) \in \mathbb{R}^K \times \mathbb{R}_{>0}^K$ and $\varepsilon \leq \epsilon_\pi \epsilon_\gamma K^{-1}$.

We show in Proposition 1 that if $o = o^*(F)$, the recursion associated to the observable (13) is optimal, and under the same condition we show in Proposition 2 that for any $\eta > 0$, the mean fields converge faster to the equilibrium $(F_1^*, \dots, F_K^*)$ (up to an additive constant) than for $\eta = 0$, as measured by the rate of decrease of a naturally associated Lyapunov function defined later in (27).

2.2.2. Estimating γ

Recall that computational resources are distributed among the rungs using the distribution $\gamma = (\gamma_1, \dots, \gamma_K)$, which is a discretization of a density originally defined over the continuous variable $\lambda \in \Lambda$. That density is often taken to be constant (and so $\gamma_k = 1/K$), but this choice is tied to the metric specified on the underlying space Λ . An optimal density must transform as a pseudotensor under coordinate transformations having Λ as their image, that is, its definition must be such that the density is multiplied by the absolute value of the Jacobian of the transformation. This requirement leads us to express γ_k as a deterministic smooth function of a collection of auxiliary averages $\mu^* = \{\mu_{km}^* = E_k[\psi_m(X)], m \in [M], k \in [K]\}$ of functions $\psi_m(x)$ against the distributions $\rho_k(x) dx$. For instance, Lindahl, Lidmar, and Hess (2018) have investigated the use of Riemannian metrics for the sampling of Λ , including the determinant of the Fisher information matrix

$$\gamma_k \propto \det \left(\left\{ \mathbb{E}_k \left[\frac{\partial H}{\partial \lambda_i} \frac{\partial H}{\partial \lambda_j} \right] - \mathbb{E}_k \left[\frac{\partial H}{\partial \lambda_i} \right] \mathbb{E}_k \left[\frac{\partial H}{\partial \lambda_j} \right] \right\}_{1 \leq i, j \leq d} \right)^{1/2}, \quad (19)$$

which we take as the default TSS reference rung density (see also Meng and Gelman 1998) and corresponds to the choice $\psi_m = \partial H / \partial \lambda_m$ for $m = 1, \dots, d$ and $\psi_m = (\partial H / \partial \lambda_i) (\partial H / \partial \lambda_j)$ for m ranging over all $\frac{d(d+1)}{2}$ pairs (i, j) such that $1 \leq i < j \leq d$. Note that for the functional dependence given by (19), there may not be any $\varepsilon > 0$ such that $\gamma(\mu) \in \mathcal{P}^\varepsilon(K)$ for all possible values of μ , so we provide a regularization in the supplementary materials).

We thus have the additional problem of estimating the unknown averages μ^* using estimates $\mu = \{\mu_{km}; m \in [M], k \in [K]\}$. Define the family of observables

$$\begin{aligned} G_{km}^\mu(x; F, \mu, o) &= \frac{1}{\pi_k} p_{\pi, F}(k|x) (\psi_m(x) - \mu_{km}) \\ &= \frac{e^{F_k - H_k(x)}}{\sum_{\ell=1}^K \pi_\ell(F, \mu, o) e^{F_\ell - H_\ell(x)}} (\psi_m(x) - \mu_{km}) \end{aligned} \quad (20)$$

with mean field $g_{km}^\mu(F, \mu, o) = \mathbb{E}_{\pi, F}[G_{km}^\mu(\mathcal{X}; F, \mu, o)]$ given by

$$\begin{aligned} g_{km}^\mu(F, \mu, o) &= \frac{1}{\pi_k} p_{\pi, F}^{[K]}(k) (\mu_{km}^* - \mu_{km}) \\ &= \frac{e^{F_k - F_k^*}}{\sum_{\ell=1}^K \pi_\ell(F, \mu, o) e^{F_\ell - F_\ell^*}} (\mu_{km}^* - \mu_{km}). \end{aligned} \quad (21)$$

Above, $\pi_\ell(F, \mu, o)$ is defined by (15) with γ_ℓ replaced by $\gamma_\ell(\mu)$. A notable property of the mean fields (21) is their *independence* from π when evaluated at $F = F^*$, so that the unique root of $g_{km}^\mu(F^*, \mu, o) = 0$ is $\mu_{km} = \mu_{km}^*$ for all $k \in [K]$, $m \in [M]$ and $o \in R_{>0}^K$.

2.2.3. Self-Adjustment

The most computationally expensive aspect of the algorithm is the iteration of the transition kernel $P_{\gamma, F} = P_{\gamma, F}^x \circ P_{\gamma, F}^k$. We anticipate this computation will be carried out on dedicated, high-performance computing units, while in parallel the free energy estimates are updated and stored on separate units.

In principle, we would like to carry out free energy updates after every iteration of $P_{\gamma, F}$, but the associated communication latency limits the frequency at which the free energies can be updated.

A partial solution is to iterate the transition kernel $P_{\gamma, F}$ multiple times (say, $\nu \geq 1$ times) between updates of the free energies, without using the $\nu - 1$ intermediate samples for said free energy update. The parameter ν controls the extent to which we sample from the target F -dependent density $p_{\gamma, F}$, and the hope is that generating samples closer to $p_{\gamma, F}$ leads to better estimates.

In his paper, Tan (2017) made an informal observation that recursion (7) used with the first marginal density in (3) has a self-adjusting mechanism, suggesting that the partial solution above is valuable. Here, we define self-adjustment as the variance reduction in the free energy estimates obtained by sampling directly from $p_{\gamma, F}$ (which corresponds to $\nu = \infty$), rather than $P_{\gamma, F}$ (which corresponds to $\nu = 1$). Below, we will quantify this variance reduction as a function of ν for a toy model.

Let $P_{\gamma, F}^\nu = P_{\gamma, F} \circ \dots \circ P_{\gamma, F}$ denote the ν th-step transition kernel, with $P_{\gamma, F} = P_{\gamma, F}^x \circ P_{\gamma, F}^k$ defined in (4)–(5). By the assumed uniform ergodicity of $P_{\gamma, F}$, $\lim_{\nu \rightarrow \infty} P_{\gamma, F}^\nu(\cdot|x, k) = p_{\gamma, F}(\cdot)$ for any (x, k) . At time t , we use $P_{\gamma, F}^\nu$ to generate a new sample $(\mathcal{X}^{t+1}, \mathcal{K}^{t+1})$ and use it to update the free energy estimates F^t .

Now, we define the toy model. Consider two overlapping uniform distributions of equal width (supplementary materials Figure 1), defined by $H_1(x) = \infty \cdot 1_{[-1+\delta, \delta]^c}(x)$ and $H_2(x) = \infty \cdot 1_{[-\delta, 1-\delta]^c}(x)$ for some $0 < \delta < 1/2$, where A^c denotes the complement of a set $A \subset \mathbb{R}$. Both uniform distributions have a width of 1, so $F_i^* = 0$ for $i = 1, 2$, and overlap on a region of width 2δ . We use $\gamma_1 = \gamma_2 = \frac{1}{2}$, $\pi = \gamma$ (no visit control) and we rewrite the estimator (7) in terms of $\Delta^t = F_2^t - F_1^t$, which yields

$$\Delta^{t+1} = \Delta^t + \frac{1}{t+1} \cdot \frac{e^{-H_1(\mathcal{X}^{t+1})} - e^{-H_2(\mathcal{X}^{t+1})}}{\frac{1}{2}e^{-H_1(\mathcal{X}^{t+1})} + \frac{1}{2}e^{-H_2(\mathcal{X}^{t+1})}}. \quad (22)$$

Assume that we can draw independent samples in x , that is, $P_{\gamma, F}^x(x'', k''|x', k') = \delta(k'' - k') \rho_{k'}(x'')$, so that any slowdown in sampling is due to the mixing properties in k . Larger ν means greater mixing, and our interest is in the variance of Δ^t as a function of ν . At time $t \geq 0$, we will use the transition kernel $P_{\gamma, F}^\nu$ to generate $(\mathcal{X}^{t+1}, \mathcal{K}^{t+1})$, where $F^t = (0, \Delta^t)$. As shown in Section 4.3.1 of the supplementary materials, the asymptotic variance of the TSS estimator is

$$\Sigma_{\text{TSS}}^\nu = 4(1 - 2\delta) \left(1 + 2 \frac{(1 - 2\delta)^\nu}{1 - (1 - 2\delta)^\nu} \right) = 4p_\delta + 8 \frac{p_\delta^{\nu+1}}{1 - p_\delta^\nu}. \quad (23)$$

Observe that Σ_{TSS}^ν has one term that decays exponentially in ν . This exponential dependence quantifies the variance reduction due to self-adjustment.

It is natural to compare the TSS estimator to the maximum likelihood estimator, often called MBAR in the chemical physics literature. The asymptotic variance of the MBAR estimator is $\Sigma_{\text{MBAR}} = 2(1 - 2\delta)/\delta = 2p_\delta/\delta$, where $p_\delta = (1 - 2\delta)$ (see Section 4.3.1 of the supplementary materials), which is independent of $\nu \geq 1$. Critically, $\Sigma_{\text{TSS}}^\nu < \Sigma_{\text{MBAR}}$ whenever $\nu > 1$ for this example. The statement that $\Sigma_{\text{TSS}}^\nu < \Sigma_{\text{MBAR}}$ for $\nu \geq 2$ requires an adequate interpretation of the covariance matrices Σ_{TSS}^ν and

Σ_{MBAR} . For an estimator such as $E^t = t^{-1} \sum_{s=1}^t \xi^s$, we define the asymptotic variance to be $\Sigma = \lim_{t \rightarrow \infty} t \sigma(t)^2$ (provided the limit exists), where $\sigma(t)^2 = \text{Var}(E^t)$. Here, we scale $\sigma(t)^2$ by the number t of samples ξ^s used in the estimator, regardless of how the samples are generated. At time $t \geq 0$, both estimators have been provided with t samples, each of which was generated using the transition kernel P_{γ, F^t}^v .

As discussed in the second part of the article, which treats the implementation of the algorithm, there is a greater computational cost associated to performing an estimator update than to making a rung move with $P_{\gamma, F}^k$: both require energy evaluations $H_k(x)$, but updating the estimators incurs computational cost due mainly to communication over an external network. The exponential dependence of the variance on v highlights the importance of distinguishing the cost of *sampling* and *estimation*, and underlines the need for transition kernels that can further improve mixing of the rung process.

2.2.4. Stochastic Approximation Triple and Theoretical Results

Here we specify the stochastic approximation procedure used by TSS and provide theoretical results on optimality and convergence. The parameter space of $\theta = (F, \mu, o)$ is $\Theta = \mathbb{R}^K \times \mathbb{R}^{KM} \times \mathbb{R}_{\geq 0}^K$, and the state space is $\mathbb{Y} = \mathcal{S} \times \mathbf{A}$, with $\mathbb{Y}^t = (\mathcal{X}^t, \mathcal{K}^t)$. In Sections 1 and 2 of the supplementary materials we use a different set of coordinates $(Z, \xi, o) \equiv (e^{-F}, e^{-F} \mu, o)$ for deriving the stochastic approximation equations presented below, because they lead to better computational behavior (i.e., (24) implies (25) has zero variance when $\psi_m(x)$ is a constant function), and because the formulation in terms of Z has a natural interpretation in terms of the Strong Law of Large Numbers (see supplementary materials Section 1.2 on Iterative Importance Sampling).

For each $k \in [K]$ and $m \in [M]$, the TSS recursions are given by

$$F_k^{t+1} = F_k^t - \log \left(1 + \frac{1}{t+1} \left[\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell=1}^K \pi_\ell(\theta^t) e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} - 1 \right] \right), \quad (24)$$

$$\mu_{km}^{t+1} = e^{F_k^{t+1} - F_k^t} \left[\mu_{km}^t + \frac{1}{t+1} \left(\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell=1}^K \pi_\ell(\theta^t) e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} \psi_m(\mathcal{X}^{t+1}) - \mu_{km}^t \right) \right], \quad (25)$$

$$o_k^{t+1} = o_k^t + \frac{1}{t+1} \left(\frac{1}{\gamma_k(\mu^t)} 1_{[k]}(\mathcal{K}^{t+1}) - o_k^t \right). \quad (26)$$

We can now describe the general TSS algorithm.

Algorithm 1

- 0) Fix $(\mathcal{X}^0, \mathcal{K}^0) \in \mathcal{S} \times [K]$ and any $\theta^0 \in \Theta$.
- 1) For $t \geq 0$, generate $\mathcal{K}^{t+1}$ from $P_{\theta^t}^k(\cdot | \mathcal{X}^t)$, and then $\mathcal{X}^{t+1}$ from $P_{\theta^t}^x(\cdot | \mathcal{K}^{t+1})$.
- 2) Update $\theta^t = (F^t, \mu^t, o^t)$ to θ^{t+1} using (24)–(26) and go to step 1.

The main object we use to establish a convergence theorem for Algorithm 1 is the relative entropy

$$V_\gamma(F) = D_{\text{KL}}(\gamma || r) = - \sum_{k \in [K]} \gamma_k \log \left(\frac{r_k(F)}{\gamma_k} \right), \quad (27)$$

with

$$r_k(F) = \gamma_k e^{F_k - F_k^*} / \left(\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*} \right). \quad (28)$$

The relative entropy is defined for any $\gamma \in \mathcal{P}^\varepsilon(K)$ and all $F \in \mathbb{R}^K$; $V_\gamma(F)$ is non-negative and equal to zero only when $F_k = F_k^* + c$ for some common additive constant $c \in \mathbb{R}$. This function serves as a global Lyapunov function for the mean field for F uniformly in $\pi \in \mathcal{P}^\varepsilon(K)$, which allows us to establish convergence of the system $\dot{F}_k(t) = g_k^F(F(t))$, $k \in [K]$ defined by (8) with γ replaced by $\pi(\theta)$ (see supplementary materials, Part 1, Section 5).

Below, we present theoretical results. The proof of Proposition 1 can be found in Section 2.2 of the supplementary materials, while the proofs of Propositions 2 and 3, which relate to visit control and self-adjustment, respectively, can be found in Sections 3.2 and 4.5 of the supplementary materials, respectively. See Section 5 of the supplementary materials for the proof of the convergence result and Theorem 1 in Section 5.5 of the supplementary materials for a general statement.

Proposition 1. The duple of recursions defined by (24) and (25) has the minimal asymptotic variance among all gain matrices, regardless of the rung-sampling probability π . If in addition $\pi = \pi^\circ$, then the triple formed by adjoining (26) also has minimal asymptotic variance.

Proposition 2. Fix an arbitrary rung density $\gamma : \mathbb{R}^{KM} \rightarrow \mathcal{P}^\varepsilon(K)$ and let $o^*(F, \mu)$ denote the steady state of the mean field of o for fixed F, μ and any visit control parameter $\eta \geq 0$. Let $F(t)$ denote the solution of the ODE $\dot{F}(t) = g_k^F(F(t), \mu(t), o^*(F(t), \mu(t)))$, where $g_k^F(F, \mu, o)$ is equal to $g_k^F(F)$ as defined in (8) with γ replaced by $\pi^{\text{TSS}}(F, \mu, o)$. Then, with $V_\gamma(F)$ defined in (27),

$$\left. \frac{d}{dt} V_\gamma(F) \right|_{\eta'} \leq \left. \frac{d}{dt} V_\gamma(F) \right|_{\eta} \leq 0, \forall t \geq 0, \eta' > \eta \geq 0. \quad (29)$$

Both inequalities are strict unless F is a constant additive shift away from F^* , in which case the time derivatives are equal to 0.

Proposition 3. Let $\pi \in \mathcal{P}^e(K)$, and suppose MBAR uses a proportion $\pi_k > 0$ of independent samples from $\rho_k(x) dx$, whereas for each fixed $F \in \mathbb{R}^K$ and the same π , TSS uses independent samples from $p_{\pi,F}(x, k) dx$. Then, for any $i, j \in [K]$, the asymptotic variance of $F_i - F_j$ for TSS is smaller than that of MBAR.

3. Implementation

Despite the theoretical features laid out in Section 2, TSS is not yet able to handle the demands of many problems of interest to the chemical or biological physicist, in which the distributions $\rho_\lambda(x) dx$ can be very dissimilar on the extent of the space Λ . Visit control is only a partial solution in such situations, because it was derived in the regime where the steady state for the rung process is quickly attained, while in practical problems the rung process $\mathcal{K}^t$ may be slaved to $\mathcal{X}^t$, leaving the rung process $\mathcal{K}^t$ stuck in a neighborhood of some $\lambda' \in \Lambda$. For λ'' far from λ' , $\rho_{\lambda'}$ and $\rho_{\lambda''}$ will often have poor overlap, which will lead to poor free energy estimates in a neighborhood of λ'' .

Our solution is to enforce the principle of locality: due to the assumed smoothness in $\lambda \rightarrow H_\lambda(x)$, ρ_λ and $\rho_{\lambda'}$ must have good overlap for λ sufficiently close to λ' . By restricting the recursion (24) to only those $k \in [K]$ with λ_k within a neighborhood of λ' , and to only those samples $(\mathcal{X}^t, \mathcal{K}^t)$ for which $\lambda_{\mathcal{K}^t}$ is in the said neighborhood, we can guarantee convergence of the free energy estimate within that neighborhood. An added benefit is that we do not waste compute evaluating $H_k(x)$ for those λ_k which are far from λ .

We describe a system of partitioning the space Λ , called windowing, to enforce this locality. In Part 2 of the supplementary materials, we describe additional practical features of the algorithm. In particular, we address the issue of the sampler's dependence on initial conditions using a version of what is known as "burn-in" suitably modified for use in on-the-fly estimators; and we modify the algorithm to use multiple replicas (i.e., multiple parallel samplers that contribute to the same free energy estimate) to take advantage of distributed computational resources. In supplementary materials Section 9, we also provide guidance on how the different features fit together.

3.1. The Windowing System

The windowing system decomposes the parameter space Λ into a collection $\{W_j; 1 \leq j \leq J\}$ of connected overlapping windows $W_j \subset \Lambda$, and separate TSS iterations are performed within each window. To link the windows together, we introduce window dynamics to move between windows, and we will thus now sample a triple $(\mathcal{X}, \mathcal{K}, \mathcal{J})$, with the specification that the stationary density conditioned on $\mathcal{J} = j$ is equal to the original stationary density $p_{\pi,F}(x, k)$ defined in (2). We provide a more detailed description for the sake of concreteness, but other variants are possible. The important attributes that must be respected are that the windows cover the entire parameter space Λ and that the window dynamics are positively recurrent.

The windows are structured so that each $\lambda \in \Lambda$ belongs to exactly two windows, with the additional constraint that the symmetric matrix $\mathcal{W}$ with the i, j entry equal to 1 if $W_i \cap W_j \neq \emptyset$ and 0 otherwise be irreducible (supplementary materials Figure 2). If we define the set $\text{win}(k) = \{j \in [J] : \lambda_k \in W_j\}$, it follows that the cardinality of $\text{win}(k)$ is 2 for each rung $k \in [K]$. For each $j \in [J]$, $F_{j\cdot}$ and $\pi_{j\cdot}$ denote the sets of per-window quantities $\{F_{j,k}; k \in [K] : \lambda_k \in W_j\}$ and $\{\pi_{j,k}; k \in [K]\}$, respectively, while the absence of a subscript denotes the totality of components over all windows: $F \equiv \{F_{j,k}; j \in [J], k : \lambda_k \in W_j\}$, for example. All per-window rung densities $\pi_{j\cdot}$ are defined so that $\pi_{j,k} \geq \varepsilon > 0$ for all $\lambda_k \in W_j$ and 0 for $\lambda_k \notin W_j$. The densities are also normalized, and thus $\sum_{k \in [K]} \pi_{j,k} = \sum_{k : \lambda_k \in W_j} \pi_{j,k} = 1$.

Next, we define the window process $\mathcal{J}^t$, which at any fixed time $t \geq 0$ is equal to one of the two possible indices in $\text{win}(\mathcal{K}^{t-1})$. The window $W_{\mathcal{J}^t}$ will be called the *active window*. The dynamics of $\mathcal{J}^t$ are determined by the transition density

$$\bar{P}_{\pi,F}^j(x', k' | x, k, j) = \delta(x - x') \delta(k - k') \times (1 - 1_{\{j\}}(j)) 1_{\text{win}(k)}(j) \quad (30)$$

which, given k , selects the other window in $\text{win}(k)$ with probability 1. This antithetic selection can of course be generalized to a random selection in which the other window is selected with probability $p \in (0, 1]$, but the need for efficient rung sampling (as discussed throughout Section 2.2) motivates us to take the most aggressive choice $p = 1$. The transition density for the triple $(\mathcal{X}, \mathcal{K}, \mathcal{J})$ is defined by $\bar{P}_{\pi,F} = \bar{P}_{\pi,F}^k \circ \bar{P}_{\pi,F}^j \circ \bar{P}_{\pi,F}^x$, where

$$\bar{P}_{\pi,F}^k(x', k' | x, k, j) \propto \delta(j' - j) 1_{W_j}(\lambda_{k'}) P_{\pi_{j\cdot}, F_{j\cdot}}^k(x', k' | x, k) \quad (31)$$

has been restricted to only propose rungs within the active window, and $\bar{P}_{\pi,F}^x(x', k' | x, k, j) = \delta(j' - j) P_{\pi,F}^x(x', k' | x, k)$ is unchanged. These dynamics and the irreducibility condition on the symmetric matrix $\mathcal{W}$ ensure each window will be visited asymptotically, and together with (30) determine the parametric invariant density to take the form

$$\bar{p}_{\pi,F}(x, k, j) = \bar{p}_{\pi,F}(j | x, k) \bar{p}_{\pi,F}(x, k) = \frac{1}{2} 1_{W_j}(\lambda_k) \bar{p}_{\pi,F}(x, k), \quad (32)$$

where $\bar{p}_{\pi,F}(j | x, k)$ denotes a conditional density and $\bar{p}_{\pi,F}(x, k)$ denotes a marginal density. In supplementary materials Section 6.1, we derive an explicit dependence of $\bar{p}_{\pi,F}(x, k)$ on πF .

To complete the description of the windowing dynamics, we must define global visit control free energies $F^\circ \equiv \{F_k^\circ; k \in [K]\}$ to be used for sampling. By analogy with the single-window setup described in Section 2.2 we impose the stationarity condition $g_{\pi_{j\cdot}, F_{j\cdot}}^\circ = 0, j \in [J]$, which relates the tilts o and density γ to the density π and visit control free energies F° . We assume that the tuple $(\mathcal{X}, \mathcal{K}, \mathcal{J})$ is drawn from the sampling density $\bar{p}_{\pi,F}(x, k, j)$ defined in (32) for fixed π, F , and that the tilts $o_{j\cdot}$ are at their steady state for each $j \in [J]$. We apply the stationarity condition to the marginal over all windows to obtain the equation

$$\sum_{j=1}^J p_j \frac{\gamma_{j,k}(\mu) o_{j,k}}{\sum_{\ell: \lambda_\ell \in W_j} \gamma_{j,\ell}(\mu) o_{j,\ell}}$$

$$= \sum_{j=1}^J p_j \frac{\pi_{j;k} e^{F_{jk} - F_k^\circ}}{\sum_{\ell: \lambda_\ell \in W_j} \pi_{j;\ell} e^{F_{j;\ell} - F_k^\circ}}, k \in [K]. \quad (33)$$

with to-be-determined marginal rung probabilities p_j , $j \in [J]$ and global visit control free energies F_k° , $k \in [K]$. As in the single-window case, when o is at its steady state for any fixed πF , the relation (33) implies $F_k^\circ = F_k^*$; otherwise, (33) defines the visit control free energy F_k° up to an additive constant. The equation itself restates the conditional probability decomposition $\tilde{\mathbb{P}}_{\pi, F}(\mathcal{K} = k) = \sum_{j \in [J]} \tilde{\mathbb{P}}_{\pi, F}(\mathcal{K} = k | \mathcal{J} = j) \tilde{\mathbb{P}}_{\pi, F}(\mathcal{J} = j)$. The quantities $F_{j;\cdot}$, $\mu_{j;\cdot}$, $o_{j;\cdot}$ are determined by Algorithm 1 restricted to window W_j . TSS is therefore a collection of weakly coupled dynamics, with the coupling being realized through F° , which enters the definition of $\pi_{j;\cdot}$ through the multi-window analogue of (16).

We note that (33) is dependent upon the tilts $o_{j;\cdot}$, which are estimated by counting events. Counting estimators are notoriously noisy, and thus the visit control free energies F_k° are only used for sampling. In the single window case, the estimated free energies F_k^t , $k \in [K]$ are directly reported back to the user. The analogue of these reported free energies in the windowed setup is obtained by letting $\eta \rightarrow \infty$. Full details relating to the implementation of (33) and the free energies reported by TSS are available in the Supplementary Materials, Part 2, Section 7.2.

4. Numerical Example

The numerical example is a toy model that illustrates key difficulties encountered in chemical physics applications, while remaining tractable enough for direct investigation. We consider the family

$$H_\lambda(x) = \frac{1}{2\sigma^2} (x - \lambda)^2 + \frac{1}{2} \log(2\pi\sigma^2) \quad (34)$$

with $\mathcal{S} = \mathbb{R}$ and $\mathbf{A} = [0, L]$. For fixed λ and variable σ , the density $\rho_\lambda(x)$, regarded as a distribution, converges to the Dirac delta about λ , in the limit $\sigma \rightarrow 0$. As $\sigma \rightarrow 0$, the number of rungs K must increase proportionally to L/σ to guarantee overlap between adjacent Gaussians. We note that, in umbrella sampling (Torrie and Valleau 1977), such approximations of the Dirac delta are used to build free energy profiles by varying λ for a fixed but sufficiently small $\sigma > 0$, which is chosen based on the resolution required for the problem.

We take $K = \lceil L/\sigma \rceil$ and study the behavior of the free energy estimates $F_K - F_1$ as $K \rightarrow \infty$; a simple rescaling argument shows that the behavior of the estimator errors is the same as for $\sigma = 1$ and $K = L + 1$. We discretize $\mathbf{A}$ by $\lambda_k = k - 1$ for $k = 1, \dots, K = L + 1$. For each $k \in [K]$, the free energy is

$$F_k^* = -\log \int_{\mathbb{R}} e^{-H_k(x)} dx = 0. \quad (35)$$

We use a flat asymptotic distribution $\gamma_k = (K - 1)^{-1}$ for $k \in \{2, \dots, K - 1\}$ and $\gamma_1 = \gamma_K = (2K - 2)^{-1}$. The sampling in x will be carried out independently given k .

4.1. Visit Control

Visit control alleviates the exponential slowdown described at the beginning of Section 2.2.1, wherein exponentially many

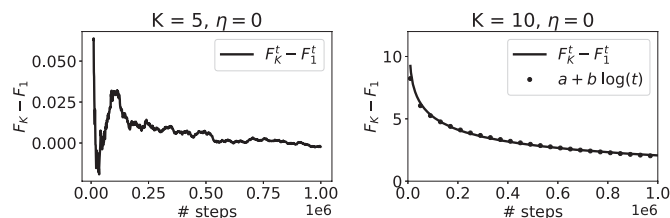


Figure 1. Free energy difference estimates $F_K - F_1$ for the Gaussian numerical example as a function of the number of steps. For $K = 5$, the estimates make it to a neighborhood of the equilibrium value 0 within 10^6 steps, while for $K = 10$ they do not. For $K = 10$, the parameters (a, b) are $(20.71, -1.35)$.

steps are required to overcome the arbitrary choice of initial estimates. Recall that the parameter $\eta \in [0, \infty)$ modulates the strength of visit control in an increasing fashion, with $\eta = 0$ corresponding to no visit control.

We first study TSS without visit control ($\eta = 0$), using a single move per estimator update ($v = 1$), and with a single window. In such a case, TSS reduces to SAMS, with the sole difference that TSS uses a logarithmic rather than a linear form of the estimator update; this difference is minor, and the algorithms are asymptotically equivalent (see supplementary materials Section 1.2 for details).

The exponential slowdown that occurs for $\eta = 0$ is a problem even for moderate values of K . To illustrate, we compare $K = 5$ with $K = 10$, running for $T = 10^6$ steps. While the estimates do reach a neighborhood of the true value of 0 for $K = 5$, they fail to do so for $K = 10$. Consistent with the observation about the exponential slowdown at the beginning of Section 2.2.1, we fit the estimates $F_K^t - F_1^t$ to the form $a + b \log t$ in this pre-asymptotic regime, demonstrating this exponential slowdown (see Figure 1).

With $\eta = 2$, which is our recommended value of η for practical simulations, and using the same number of steps as previously, we are able to recover convergence for K as large as 50 (see Figure 2, left panel).

Larger values of K demonstrate again the existence of a slowdown, as illustrated in the right panel of Figure 2 for $K = 100$, but we suspect that the rate at which the free energy estimates approach the asymptotic regime is a law of the form $\propto t^{-\beta}$. The inverse power law makes it feasible to attain the asymptotic regime by increasing the number of samples, as opposed to the case of no visit control in which such an approach is hopeless because of the exponential slowdown. However, the windowing system provides a better alternative, one that scales with the size of the problem.

4.2. Windows

In the previous example we increased K , thereby increasing the difficulty of the problem, but we did not correspondingly increase the simulation time, so as to have kept the number of visits to each rung constant between simulations. The reason is that with a constant simulation time, the number of energy evaluations $H_k(x)$ scales linearly with the size of the problem, that is, K . It is thus an inherent limitation of single-window TSS that scaling the number of samples with the size of the problem creates a quadratic scaling in the number of energy evaluations.

Windows provide a method of scaling the number of energy evaluations linearly with the size of the system while keeping the

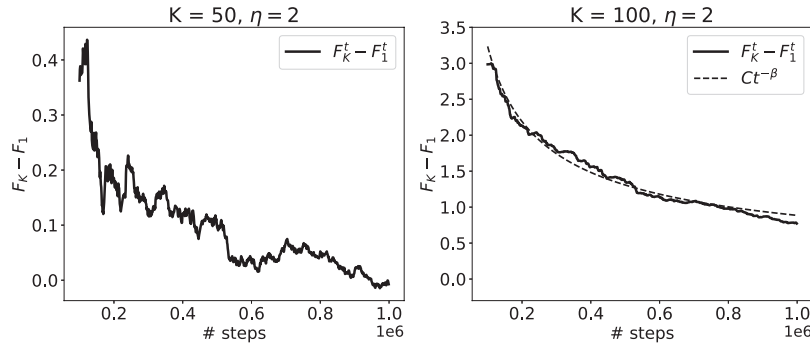


Figure 2. Free energy difference estimates $F_K - F_1$ for the Gaussian numerical example as a function of the number of steps. With visit control, in particular using $\eta = 2$, the estimates make it to a neighborhood of the equilibrium value 0 within 10^6 steps even for $K=50$, and do not suffer from the exponential slowdown illustrated in Figure 1. $K=100$ illustrates, however, that a slowdown still exists, though it may be in the form of a power law; here, we fit $F_K^t - F_1^t = Ct^{-\beta}$, where $(C, \beta) = (2108, 0.56)$.

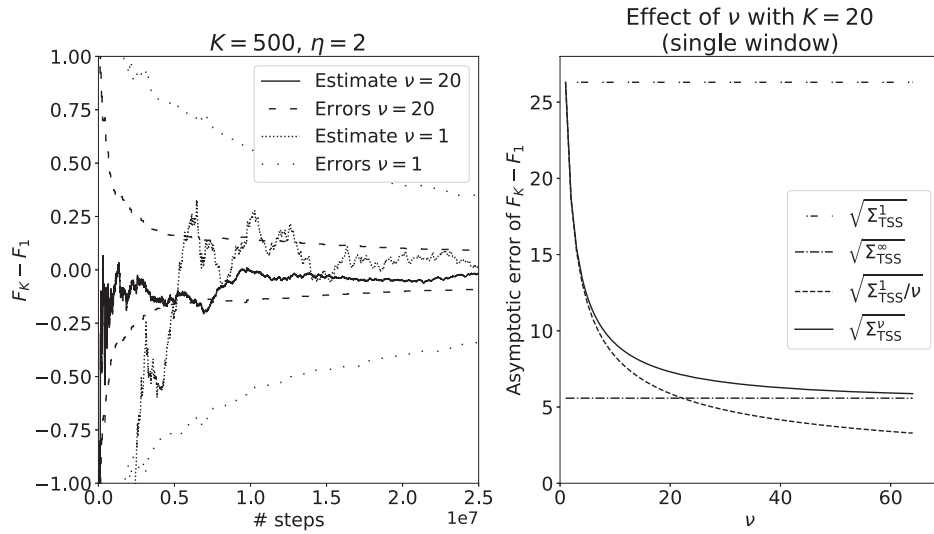


Figure 3. (Left) free energy difference estimates for the Gaussian numerical example with $K=500$ using 51 windows (two windows of size 10 covering rungs 0–9 and 490–499, and windows of size 20 of the form $[20i, 20(i+1))$ for $i = 0, \dots, 24$ and $[10 + 20i, 10 + 20(i+1))$ for $i = 0, \dots, 23$). Free energy estimates are shown for $\nu = 1$ (dotted) and $\nu = 20$ (solid), along with twice their estimated errors (dashed for $\nu = 20$ and sparsely dotted for $\nu = 1$). (ν is the number of subdivisions per TSS cycle; see Section 4.3.) (Right) Analytical values of the asymptotic error for a single-window simulation as a function of ν (see Section 4.3).

number of samples per rung constant. In Figure 3 (left panel), we scale the case of $(K, T) = (50, 10^6)$, which had 5×10^7 energy evaluations, to $(K, T) = (500, 2.5 \times 10^7)$ with windows of size 20, which uses roughly $20 \times 2.5 \times 10^7 = 5 \times 10^8$ energy evaluations. The results illustrate how windows and visit control together enable TSS to adapt to arbitrarily large values of K . This is important for many practical applications, particularly those pertaining to the estimation of free energy profiles, which involve very large numbers of rungs with widely dissimilar distributions.

We note that in general smaller windows increase the rate at which the TSS estimates approach the asymptotic regime, though at the price of limiting the rung movement, with the consequence of larger asymptotic error bars. Windows therefore allow the user to increase the rate at which TSS enters the asymptotic regime by trading it for larger asymptotic error bars, with the tradeoff optimized differently on different regions of Λ .

4.3. Self-Adjustment

Although use of windows eliminates the quadratic scaling of the number of energy evaluations with K , the estimation still

requires solving a convex optimization problem in which the windows are weakly coupled, and whose cost scales linearly with K asymptotically. On today's heterogeneous computational architectures, the energy evaluations are performed on specialized processors such as GPUs or Anton (Shaw et al. 2021; Bergdorf et al. 2021) ASICs (the “sampler”), while the optimization problem is solved on a cluster of highly programmable processors (the “estimator”). One thus incurs the cost of communicating between the two, and it becomes desirable to find ways of hiding this communication so as to allow the sampler to continue without having to wait for the estimator.

Figure 4, which illustrates a full TSS cycle, shows that the generation of $\mathcal{X}$ samples can be carried out while waiting for an estimator update. Once the values of F are received, we can proceed with the dynamics in $\mathcal{K}$. For our Gaussian example, the $\mathcal{X}$ samples are conditionally independent given $\mathcal{K}$, and particularly fast. What else is there to do while the sampler waits for the estimator values to arrive? Drawing a new sample $\mathcal{X}$ from the same $\mathcal{K}$ brings no advantage, but the distribution of $(\mathcal{X}, \mathcal{K})$ can be brought closer to $p_F(x, k)$ by iterating the composed sampler $P_F = P_F^x \circ P_F^k$ a number of ν times using the most recent free energy estimates F available to the sampler.

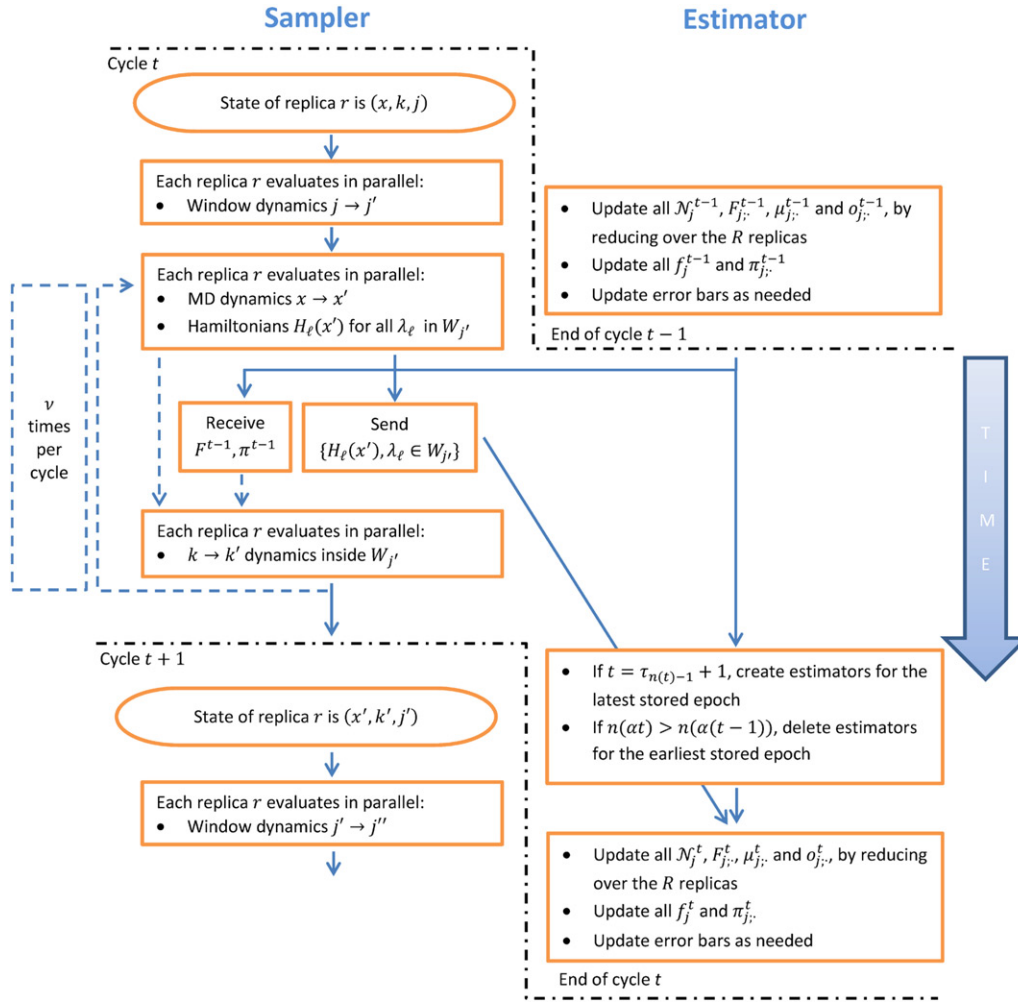


Figure 4. A full TSS cycle. The sampling moves (processes in the left column) involve energy and force evaluations, and the estimator updates (processes in the right column) perform the per-window stochastic approximation updates and produce global free energy estimates. The sampling and estimation can be executed on separate hardware devices, and both occur during a TSS cycle, the start and end of which is delineated between black dash-dotted lines. When the result of an operation is transmitted exactly once per TSS cycle, this is shown with a solid blue line; when the result may be transmitted more frequently, this is shown with a dashed blue line. The sampler device receives updated estimates (F^{t-1}, π^{t-1}) from the estimator device once per cycle, and these received estimates are used for the first time in a cycle only after the completion of the first MD dynamics and Hamiltonian evaluations. These Hamiltonian evaluations (evaluated at the end of this first run of MD dynamics) are sent to the estimator device for further updates. The MD dynamics, Hamiltonian evaluations, and rung move ($k \rightarrow k'$ dynamics) are iterated an additional $\nu - 1$ times within a cycle, with the same estimate (F^{t-1}, π^{t-1}) used for the rung moves in all $\nu - 1$ additional steps; these additional Hamiltonian evaluations are not sent to the estimator device. At the end of a cycle, the x and k dynamics have been updated in an interleaving fashion a total of ν times. While the additional $\nu - 1$ iterations of (x, k) dynamics are performed for each replica by the sampler device, the estimator device updates the epoch counts, receives the Hamiltonian evaluations, uses them to compute the updated global estimates, and sends the updated estimates to the sampler device in time for the first $k' \rightarrow k''$ dynamics of the following cycle. (Full details for defining the quantities referred to in the figure are available in the supplementary materials.)

Generating samples directly from $p_F(x, k)$, as the above method does in the case of $\nu \rightarrow \infty$, makes the TSS estimates better than any other known method (see Proposition 3): By expanding on the calculations in Tan's work (see supplementary materials Section 4.4), we analytically compute the asymptotic errors within a single window and plot them as a function of ν for a window of size 20 in the right panel of Figure 3, and see that the improvement can be dramatic. The left panel of Figure 3 then compares the behavior of the free energy estimates with $\nu = 1$ and $\nu = 20$ as a function of the number of estimator updates, using multiple windows of size 20. The numerical simulation provides evidence that improvements in the single-window case carry over to the multiple window case.

The analytical results compare two schemes: one in which each sample is used for an estimator update (dashed curve, $\sqrt{\Sigma_{\text{TSS}}^1} / \sqrt{\nu}$), and which thus requires communication between the sampler and the estimator at every step, and another in which only the ν th sample is used for an estimator update (solid curve, $\sqrt{\Sigma_{\text{TSS}}^\nu}$), which only requires communication every ν th step. The results show that the communication-intensive scheme (dashed curve) is statistically more efficient than the scheme using only one of the ν samples for estimation (solid curve), though notably the improvement is small for small values of ν (say < 20). In other words, the majority of the variance reduction relative to the horizontal dot-dashed curve ($\nu = 1$) is accounted for by the improved sampling of the rung process

$\mathcal{K}^t$ in both cases. This observation is particularly relevant for umbrella sampling, where the energy evaluations required for the rung move can be cheaply computed, thereby allowing us to achieve much lower error without paying the price of communication.

Supplementary Materials

The supplementary materials provide additional details on derivations and proofs, along with a numerical example that uses Times Square Sampling in the context of molecular dynamics simulation.

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Disclosure Statement

The authors declare no competing interests.

Data Availability Statement

The source code containing the Times Square sampling algorithm is available at <https://github.com/DEShawResearch/tss>.

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